

Generation Interconnection System Impact Study Report

For Queue Project Q21-1802

Gentilly Road 138 kV

July 16, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1802. The Interconnection Customer proposes a 203.7 MW solar photovoltaic facility interconnecting at the Gentilly Road 138 kV substation in the Delaware area. The request was received on June 29, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 415 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,957,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Sunset Lake – Broadview 138kV
Contingency: Denham Springs – Otay Tap 138kV
Loading: 111.8% (Rating: 195 MVA)
DFAX: 12.3%
Pre-project loading: 98.9%
2. Facility: Southpoint – Mill Cteek 138kV
Contingency: Dell Crk – Keeney Substation 138kV
Loading: 105.5% (Rating: 172 MVA)
DFAX: 8.8%
Pre-project loading: 95.1%
3. Facility: Gentilly Rd – Churchs Ferry Substation 138kV
Contingency: Chambers 345/138kV Auto
Loading: 101.6% (Rating: 201 MVA)
DFAX: 82.1%
Pre-project loading: 18.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sunset Lake – Broadview Substation 138kV	Rebuild line with higher-capacity conductor	\$3,514,000	100% (\$3,514,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mill Cteek – Southpoint 138kV	Reconductor line	\$1,095,000	100% (\$1,095,000)
Gentilly Rd – Churchs Ferry 138kV Line	Reconductor line	\$5,305,000	63% (\$3,348,000)
	Total allocated to project		\$7,957,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.